

# Dynamic Programming & Reinforcement Learning

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# Content for Today

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## Setting and notation

- Finite, nonempty state and action spaces  $\mathcal{X}, \mathcal{A}$ . The state is fully observed.
- Given the observed history and current action, the joint law of  $(R_t, X_{t+1})$  depends only on  $(X_t, A_t)$ , with the same rule at every time.
- $r(x, a)$  is the mean immediate reward;  $\mathcal{P}(x' \mid x, a)$  is a next-state probability.
- $|R_t| \leq R_{\max} < \infty$  with probability one;  $0 \leq \gamma < 1$ .

A stationary Markov policy uses  $\pi(a \mid x)$ , or a single action  $\pi(x)$  when deterministic. Later we compare with all history-dependent policies.

$\mathcal{B}(\mathcal{X})$  denotes bounded real functions on  $\mathcal{X}$ ; similarly for  $\mathcal{B}(\mathcal{X} \times \mathcal{A})$ . On finite domains, every real-valued table is bounded.

## Recap: the Bellman equations

For a fixed stationary Markov policy, the expected returns satisfy

$$V^\pi(x) = \sum_a \pi(a | x) \left[ r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' | x, a) V^\pi(x') \right],$$

$$Q^\pi(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' | x, a) \sum_{a'} \pi(a' | x') Q^\pi(x', a').$$

The optimality equations we aim to justify are

$$V^*(x) = \max_a \left[ r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' | x, a) V^*(x') \right],$$

$$Q^*(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' | x, a) \max_{a'} Q^*(x', a').$$

Do these equations have unique solutions? Why do their solutions describe optimal decisions?

## Greedy policies

Given a candidate action-value function  $Q$ , choose

$$\pi_g(x; Q) \in \operatorname{argmax}_{a \in \mathcal{A}} Q(x, a).$$

Given a candidate state-value function  $V$ , choose

$$\pi_g(x; V) \in \operatorname{argmax}_{a \in \mathcal{A}} \left[ r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \right].$$

- $\max$  gives a number;  $\operatorname{argmax}$  gives maximizing actions.
- Fix one maximizing action at each state, including when actions tie.
- Write  $\pi_g(V)$  or  $\pi_g(Q)$  for the resulting state-to-action rule.

## Why a greedy policy can plan ahead

For the state-value rule, each candidate action receives the score

$$\underbrace{r(x, a)}_{\text{reward now}} + \gamma \underbrace{\sum_{x'} \mathcal{P}(x' | x, a) V(x')}_{\text{expected continuation value}} .$$

- The action is selected before observing the next state.
- Future consequences enter through  $V$ , so we need only a one-step calculation at decision time.
- With the correct optimal values, a greedy rule is optimal. We will prove this claim.
- With an arbitrary candidate, the resulting greedy policy may be suboptimal.

# Bellman Operators



# Operators

An operator maps elements of one space to another. For fixed-point problems, we study operators  $L : \mathcal{Z} \rightarrow \mathcal{Z}$  that map a space into itself.

- On real numbers:  $L : \mathbb{R} \rightarrow \mathbb{R}$ ,  $L(z) = z^2$ , so  $L(5) = 25$ .
- On smooth functions  $f : \mathbb{R} \rightarrow \mathbb{R}$ :  $L(f) = f'$ , so  $L(\sin) = \cos$ .
- On state-value functions:  $T : \mathcal{B}(\mathcal{X}) \rightarrow \mathcal{B}(\mathcal{X})$ ,  $V \mapsto T(V)$ .

For  $V \in \mathcal{B}(\mathcal{X})$  and  $x \in \mathcal{X}$ , we write

$$\underbrace{TV := T(V)}_{\text{an output function in } \mathcal{B}(\mathcal{X})}, \quad \underbrace{(TV)(x) = (T(V))(x)}_{\text{a real number at state } x}.$$

Apply  $T$  to the whole function  $V$ , then evaluate the output at  $x$ . A candidate  $V$  is any real value table. It need not be the return of a policy.

## Bellman operators for a policy

Fix a stationary Markov policy  $\pi$ . Its mean reward is

$$r^\pi(x) = \sum_a \pi(a | x) r(x, a).$$

For any candidates  $V \in \mathcal{B}(\mathcal{X})$ ,  $Q \in \mathcal{B}(\mathcal{X} \times \mathcal{A})$ , define

$$(T^\pi V)(x) = r^\pi(x) + \gamma \sum_a \pi(a | x) \sum_{x'} \mathcal{P}(x' | x, a) V(x').$$

$$(T^\pi Q)(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' | x, a) \sum_{a'} \pi(a' | x') Q(x', a').$$

The same symbol  $T^\pi$  denotes two mappings. The input tells us whether the output is a state table or a state-action table.

## Deterministic policies

If  $\pi(x)$  specifies one action at each state, the action averages disappear:

$$(T^\pi V)(x) = r(x, \pi(x)) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, \pi(x)) V(x'),$$

$$(T^\pi Q)(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) Q(x', \pi(x')).$$

- In the first equation, the current action is  $\pi(x)$ .
- In the second, the current action  $a$  is prescribed. The policy selects the next action  $\pi(x')$ .
- A deterministic policy can still produce random next states and rewards.

# Bellman operators and Bellman equations

The return-based values satisfy

$$V^\pi = T^\pi V^\pi, \quad Q^\pi = T^\pi Q^\pi.$$

The policy Bellman equations are therefore **fixed-point equations**.

- **One update:** supply a candidate  $V$ , calculate  $T^\pi V$ .
- **A solution:** find a candidate unchanged by the update.

$$V_{k+1} = T^\pi V_k$$

Here  $k$  counts computations. It is different from the environment time  $t$ . We still need to prove uniqueness and convergence of these updates.

## Bellman optimality operators

For arbitrary candidates, define

$$(T^*V)(x) = \max_a \left[ r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \right],$$

$$(T^*Q)(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) \max_{a'} Q(x', a').$$

- For  $V$ , maximize the current action before seeing the next state.
- For  $Q$ , keep the current action fixed, then maximize the next action at each possible next state.

For every candidate  $V$ ,  $T^{\pi_g(V)}V = T^*V$ .

# The optimality fixed-point equations

The equations to solve are

$$V = T^*V, \quad Q = T^*Q.$$

We will establish:

1. These equations have unique solutions.
2. Repeated updates converge to their solutions.
3. A greedy policy attains the state-value solution.
4. No admissible policy achieves a larger expected return.

Call the state-value solution  $U$  until we prove  $U = V^*$ . Monotonicity will help compare policy values. Contraction will give unique solutions and convergent updates.

# Properties of Bellman Operators

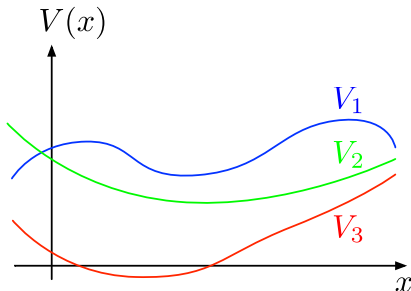
## Monotonicity



## Pointwise order

For  $V_1, V_2 \in \mathcal{B}(\mathcal{X})$ ,

$$V_1 \leq V_2 \iff V_1(x) \leq V_2(x) \text{ for every } x \in \mathcal{X}.$$



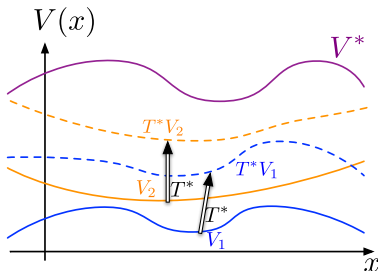
$V_3 \leq V_1$  and  $V_3 \leq V_2$ . Neither  $V_1 \leq V_2$  nor  $V_2 \leq V_1$ .

The curves illustrate functions; a finite value table uses the same comparison at every state.

## Monotonicity of Bellman operators

If  $V_1 \leq V_2$ , then

$$T^\pi V_1 \leq T^\pi V_2, \quad T^* V_1 \leq T^* V_2.$$



Compare the two solid inputs, then the two dashed outputs. Their order is preserved. This does not assert  $TV \geq V$ ; the upward arrows are specific to this illustration.

## Monotonicity: proof

Let  $V_1(x') \leq V_2(x')$  for every next state. Since probabilities and  $\gamma$  are nonnegative,

$$\begin{aligned}(T^\pi V_1)(x) &= r^\pi(x) + \gamma \sum_a \pi(a | x) \sum_{x'} \mathcal{P}(x' | x, a) V_1(x') \\ &\leq r^\pi(x) + \gamma \sum_a \pi(a | x) \sum_{x'} \mathcal{P}(x' | x, a) V_2(x') \\ &= (T^\pi V_2)(x).\end{aligned}$$

For  $T^*$ , every action's score under  $V_1$  is at most its score under  $V_2$ . Taking their maxima preserves this inequality.

The same reasoning applies to the  $Q$  operators: next-action averages and maxima both preserve pointwise order.

Rewards may have either sign. The same reward term appears on both sides.

# Metric, Norm, and Contraction Mappings



# Metric

A **metric** is a distance function  $d : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$  satisfying, for all  $x, y, z \in \mathcal{Z}$ :

1.  $d(x, y) \geq 0$ , and  $d(x, y) = 0$  exactly when  $x = y$ .
2.  $d(x, y) = d(y, x)$  (symmetry).
3.  $d(x, y) \leq d(x, z) + d(z, y)$  (triangle inequality).

A **metric space**  $(\mathcal{Z}, d)$  is a set together with a metric.

$$\mathcal{Z} = \mathbb{R}, \quad d(x, y) = |x - y|, \quad d(1, 4) = 3.$$

A metric compares two objects. They could be numbers, vectors, or functions.

# Norm

A **norm** assigns a size  $\|z\|$  to a vector in a real linear space. For all vectors  $z, w$  and scalars  $\lambda \in \mathbb{R}$ :

1.  $\|z\| \geq 0$ , and  $\|z\| = 0$  exactly when  $z = 0$ .
2.  $\|\lambda z\| = |\lambda| \|z\|$  (homogeneity).
3.  $\|z + w\| \leq \|z\| + \|w\|$  (triangle inequality).

A linear space allows addition and scalar multiplication. Function spaces can also be linear spaces.

$$\boxed{d(z, w) = \|z - w\|}$$

Every norm therefore induces a metric: measure the size of the difference.

## Norms of vectors

For  $z = (z_1, \dots, z_m) \in \mathbb{R}^m$ , commonly used norms are

$$\|z\|_p = \left( \sum_{i=1}^m |z_i|^p \right)^{1/p}, \quad 1 \leq p < \infty, \quad \|z\|_\infty = \max_i |z_i|.$$

For  $z = (3, 4)$ ,

$$\|z\|_1 = 3 + 4 = 7, \quad \|z\|_2 = \sqrt{9 + 16} = 5, \quad \|z\|_\infty = 4.$$

- Different norms measure the size of the same vector differently.
- The maximum norm controls every coordinate at once:  $|z_i| \leq \|z\|_\infty$ .

## Norms of functions

Let  $C([0, 1])$  be the space of continuous real functions on  $[0, 1]$ . The **supremum norm**, also called the **uniform norm**, is

$$\|f\|_{\infty} = \sup_{x \in [0, 1]} |f(x)|.$$

sup means the least upper bound. A continuous function on the closed, bounded interval  $[0, 1]$  attains its largest absolute value.

For  $f(x) = 2x - 1$ , we have  $-1 \leq f(x) \leq 1$ , so  $\|f\|_{\infty} = 1$ .

$$d(f, g) = \|f - g\|_{\infty} = \sup_{x \in [0, 1]} |f(x) - g(x)|.$$

This distance measures the largest vertical gap between the two functions.

## Supremum norms for value functions

For bounded value candidates,

$$\|V\|_{\infty} = \sup_{x \in \mathcal{X}} |V(x)|, \quad \|Q\|_{\infty} = \sup_{(x,a) \in \mathcal{X} \times \mathcal{A}} |Q(x,a)|.$$

On our finite domains, each supremum is a maximum. These are maximum *absolute* values, not maximum signed values.

$$V = (-10, 2)^{\top} \implies \|V\|_{\infty} = 10.$$

The distance used for Bellman operators is

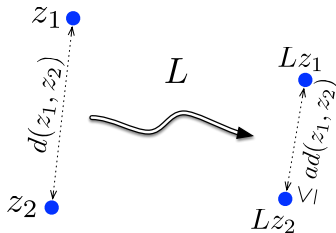
$$d(V_1, V_2) = \|V_1 - V_2\|_{\infty}.$$

Thus  $d(V_1, V_2) \leq \varepsilon$  means that the two candidates differ by at most  $\varepsilon$  at every state. The same applies to  $Q$ .

## Contraction mapping

On a metric space  $(\mathcal{Z}, d)$ , a mapping  $L : \mathcal{Z} \rightarrow \mathcal{Z}$  is a **contraction** if a constant  $0 \leq a < 1$  satisfies

$$d(Lz_1, Lz_2) \leq a d(z_1, z_2) \quad \text{for all } z_1, z_2 \in \mathcal{Z}.$$



The same factor must work for every pair. The output distance is at most that fraction of the input distance.

## A scalar contraction

Let  $\mathcal{Z} = \mathbb{R}$ ,  $d(z_1, z_2) = |z_1 - z_2|$ , and  $Lz = az$ , where  $a \in \mathbb{R}$ .

$$\begin{aligned}d(Lz_1, Lz_2) &= |az_1 - az_2| \\&= |a| |z_1 - z_2| \\&= |a| d(z_1, z_2).\end{aligned}$$

The mapping is a contraction exactly when  $|a| < 1$ , with factor  $|a|$ .

For  $a = -\frac{1}{2}$ , distances shrink even though the mapping reverses order.

Monotonicity concerns order; contraction concerns distance. These are different properties.

## Iteration of a scalar mapping

Starting at  $z_0$ , repeatedly apply  $Lz = az$ :

$$z_1 = az_0, \quad z_2 = a^2 z_0, \quad \dots, \quad z_k = a^k z_0.$$

$$|a| < 1$$

Every initial value converges to zero.

$$a = 1$$

Each sequence stays at its own initial value.

$$a = -1$$

A nonzero initial value alternates between  $z_0$  and  $-z_0$ .

$$|a| > 1$$

Magnitude grows without bound unless  $z_0 = 0$ .

Contraction guarantees convergence to the same point from every start in this example.

Convergence of a particular sequence alone does not imply contraction.

## Fixed point

A **fixed point** of  $L : \mathcal{Z} \rightarrow \mathcal{Z}$  is an element  $z^*$  with

$$Lz^* = z^*.$$

Examples:

- $Lz = \frac{1}{2}z$  has the fixed point  $z^* = 0$ .
- $Lz = z$  fixes every real number.
- $Lz = z + 1$  has no real fixed point.

Finding a fixed point and proving that iteration reaches it are separate questions.

For Bellman operators, a fixed point is an entire value function unchanged by the update.

## An equation as a fixed-point problem

To solve  $f(z) = 0$ , define

$$Lz = z + f(z).$$

Then

$$Lz^* = z^* \iff f(z^*) = 0.$$

So the fixed points are exactly the roots of the original equation.

However, this rearrangement alone does not make  $L$  a contraction. We must still check whether iteration converges.

Different rearrangements of the same equation can lead to different iteration behavior.

## Cauchy sequences and completeness

A sequence  $(z_k)$  is **Cauchy** if, for every  $\varepsilon > 0$ , there is an  $N$  such that

$$d(z_m, z_n) < \varepsilon \quad \text{whenever } m, n \geq N.$$

All sufficiently late terms are close to one another. A metric space is **complete** if every Cauchy sequence has a limit *inside the space*.

- The real numbers with absolute distance are complete.
- In  $\mathbb{R}^m$  with the maximum norm, a Cauchy sequence is Cauchy in each coordinate. Each coordinate has a real limit.
- Finitely many coordinate limits give convergence of the whole vector in the maximum norm.

Therefore our finite value-table spaces are complete.

# Banach fixed-point theorem

Let  $(\mathcal{Z}, d)$  be a *nonempty complete* metric space. If  $L : \mathcal{Z} \rightarrow \mathcal{Z}$  is a contraction with factor  $0 \leq a < 1$ , then:

1. There is exactly one fixed point  $z^*$ .
2. From any  $z_0 \in \mathcal{Z}$ , the iteration  $z_{k+1} = Lz_k$  converges to  $z^*$ .
3.  $d(z_k, z^*) \leq a^k d(z_0, z^*)$ .

**Why it works.** Contraction makes successive differences shrink geometrically. Their tail sum makes the iterates Cauchy. Completeness gives a limit; contraction makes that limit a fixed point and excludes a second one.

Detailed proof: optional slides ??-??.

## Class exercise: a linear equation

**Three minutes.** We want to solve  $cz + b = 0$ , with  $c, b \in \mathbb{R}$ .

1. Use the previous rearrangement to find  $Lz$ .
2. For which  $c$  is this mapping a contraction?
3. With  $c = -0.5$ ,  $b = 1$ ,  $z_0 = 0$ , find  $z_1, z_2$  and the fixed point.

$$Lz = \underline{\hspace{2cm}},$$

$$z_1 = \underline{\hspace{1cm}}, \quad z_2 = \underline{\hspace{1cm}}.$$

### Tools to use

$$f(z) = cz + b,$$

$$Lz = z + f(z).$$

$$z_{k+1} = Lz_k.$$

A contraction requires

$$|Lz - Lw| \leq a|z - w|,$$

with one  $0 \leq a < 1$  for all  $z, w$ .

## Class exercise: solution

The prescribed rearrangement gives

$$Lz = z + (cz + b) = (1 + c)z + b.$$

For every  $z, w$ ,

$$|Lz - Lw| = |1 + c| |z - w|.$$

Thus this mapping is a contraction exactly when

$$|1 + c| < 1 \iff \boxed{-2 < c < 0.}$$

For  $c = -0.5$ ,  $b = 1$ , the update is  $z_{k+1} = 0.5z_k + 1$ :

$$z_0 = 0, \quad z_1 = 1, \quad z_2 = 1.5, \quad z_3 = 1.75, \quad \dots \rightarrow 2.$$

The fixed point solves  $z^* = 0.5z^* + 1$ , giving  $z^* = 2$ , which also solves  $-0.5z + 1 = 0$ .

# Contraction of Bellman Operators



## Bellman operators are contractions

For every fixed stationary policy  $\pi$ , both  $T^\pi$  and  $T^*$  are  $\gamma$ -contractions in the supremum norm.

$$\|TV_1 - TV_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty,$$

$$\|TQ_1 - TQ_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty,$$

where  $T$  is the relevant policy or optimality operator.

We first prove the policy  $Q$  case. We then handle the maximum in  $T^*$ .

Probability averaging does not enlarge the largest error. Discounting multiplies the bound by  $\gamma < 1$ .

## Policy contraction: reward cancellation

Fix the same policy  $\pi$ , and any two candidates  $Q_1, Q_2$ . For every  $(x, a)$ ,

$$\begin{aligned} & (T^\pi Q_1)(x, a) - (T^\pi Q_2)(x, a) \\ &= \left[ r(x, a) + \gamma \sum_{x', a'} \mathcal{P}(x' | x, a) \pi(a' | x') Q_1(x', a') \right] \\ &\quad - \left[ r(x, a) + \gamma \sum_{x', a'} \mathcal{P}(x' | x, a) \pi(a' | x') Q_2(x', a') \right] \\ &= \gamma \sum_{x', a'} \mathcal{P}(x' | x, a) \pi(a' | x') (Q_1(x', a') - Q_2(x', a')). \end{aligned}$$

The reward cancels. The nonnegative weights sum to one:

$$\sum_{x', a'} \mathcal{P}(x' | x, a) \pi(a' | x') = \sum_{x'} \mathcal{P}(x' | x, a) \underbrace{\sum_{a'} \pi(a' | x')}_1 = 1.$$

## A probability average is bounded by the largest magnitude

Let  $p_i \geq 0$  with  $\sum_i p_i = 1$ , and let  $f_i \in \mathbb{R}$ . Then

$$\begin{aligned} \left| \sum_i p_i f_i \right| &\leq \sum_i p_i |f_i| \\ &\leq \sum_i p_i \max_j |f_j| \\ &= \max_j |f_j|. \end{aligned}$$

The first step uses the triangle inequality. The second bounds each term by the same number. The last uses total probability one.

$$\boxed{|\mathbb{E}[f(Y)]| \leq \mathbb{E}[|f(Y)|] \leq \|f\|_\infty.}$$

## Policy contraction: completing the bound

For any  $(x, a) \in \mathcal{X} \times \mathcal{A}$ , the preceding cancellation gives

$$\begin{aligned} & |(T^\pi Q_1)(x, a) - (T^\pi Q_2)(x, a)| \\ &= \gamma \left| \sum_{x', a'} \mathcal{P}(x' \mid x, a) \pi(a' \mid x') (Q_1(x', a') - Q_2(x', a')) \right| \\ &\leq \gamma \sum_{x', a'} \mathcal{P}(x' \mid x, a) \pi(a' \mid x') |Q_1(x', a') - Q_2(x', a')| \\ &\leq \gamma \|Q_1 - Q_2\|_\infty \sum_{x', a'} \mathcal{P}(x' \mid x, a) \pi(a' \mid x') = \gamma \|Q_1 - Q_2\|_\infty. \end{aligned}$$

This bound holds for **every**  $(x, a)$ , and its right-hand side is **independent of**  $(x, a)$ . Taking the maximum therefore gives

$$\|T^\pi Q_1 - T^\pi Q_2\|_\infty = \max_{x, a} |(T^\pi Q_1)(x, a) - (T^\pi Q_2)(x, a)| \leq \gamma \|Q_1 - Q_2\|_\infty.$$

## Policy contraction: state values

For every state  $x$ , reward cancellation and the same averaging bound give

$$\begin{aligned} & |(T^\pi V_1)(x) - (T^\pi V_2)(x)| \\ &= \gamma \left| \sum_{a, x'} \pi(a | x) \mathcal{P}(x' | x, a) (V_1(x') - V_2(x')) \right| \\ &\leq \gamma \sum_{a, x'} \pi(a | x) \mathcal{P}(x' | x, a) |V_1(x') - V_2(x')| \\ &\leq \gamma \|V_1 - V_2\|_\infty \underbrace{\sum_a \pi(a | x)}_1 \underbrace{\sum_{x'} \mathcal{P}(x' | x, a)}_1 = \gamma \|V_1 - V_2\|_\infty. \end{aligned}$$

The right-hand side is independent of  $x$ . Taking the maximum over states gives

$$\boxed{\|T^\pi V_1 - T^\pi V_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty.}$$

## A bound for two maxima

For two lists  $(u_a)$  and  $(w_a)$  on the same finite nonempty action set,

$$\left| \max_a u_a - \max_a w_a \right| \leq \max_a |u_a - w_a|.$$

Let  $D = \max_a |u_a - w_a|$ . For each  $a$ ,

$$u_a \leq w_a + D \leq \max_b w_b + D.$$

Taking the maximum over  $a$  gives

$$\max_a u_a - \max_a w_a \leq D.$$

Exchange the lists to obtain the reverse bound.

The two lists may have different maximizing actions. This proof does not assume they share a maximizer.

## Contraction of the optimality operators

Apply the maximum inequality to the action scores in  $T^*V$ :

$$\begin{aligned} |(T^*V_1)(x) - (T^*V_2)(x)| &\leq \gamma \max_a \left| \sum_{x'} \mathcal{P}(x' | x, a) (V_1(x') - V_2(x')) \right| \\ &\leq \gamma \|V_1 - V_2\|_\infty. \end{aligned}$$

For  $Q$ , first compare the next-action maxima:

$$\left| \max_{a'} Q_1(x', a') - \max_{a'} Q_2(x', a') \right| \leq \|Q_1 - Q_2\|_\infty.$$

Then the transition average gives

$$|(T^*Q_1)(x, a) - (T^*Q_2)(x, a)| \leq \gamma \sum_{x'} \mathcal{P}(x' | x, a) \|Q_1 - Q_2\|_\infty = \gamma \|Q_1 - Q_2\|_\infty.$$

Maximize over the input entries. Both optimality operators are  $\gamma$ -contractions.

# Fixed Points, Greedy Policies, and Error Bounds



## Existence, uniqueness, and convergence

The finite value-table spaces, with supremum-norm distance, are nonempty and complete. Each Bellman operator maps its space into itself and is a  $\gamma$ -contraction.

By Banach's theorem, each of the four operators has exactly one fixed point. From any initial candidate, its iterates converge to that fixed point.

$$V_{k+1} = T^\pi V_k \longrightarrow V^\pi, \quad V_{k+1} = T^* V_k \longrightarrow U.$$

The policy fixed point is  $V^\pi$  because return decomposition already gives  $V^\pi = T^\pi V^\pi$ . Likewise, the policy  $Q$  fixed point is  $Q^\pi$ . For the optimality iteration  $V_{k+1} = T^* V_k$ ,

$$\|V_k - U\|_\infty \leq \gamma^k \|V_0 - U\|_\infty.$$

Here  $U = T^* U$ . Identifying  $U$  as the optimal return still requires proof.

## When a policy attains the fixed point

Let  $U = T^*U$ . For a stationary Markov policy  $\pi$ ,

$$\boxed{T^\pi U = T^*U \iff V^\pi = U.}$$

**Forward.** If  $T^\pi U = T^*U = U$ , then  $U$  is a fixed point of  $T^\pi$ . Its unique fixed point is  $V^\pi$ , so  $V^\pi = U$ .

**Reverse.** If  $V^\pi = U$ , then

$$T^\pi U = T^\pi V^\pi = V^\pi = U = T^*U.$$

This tells us when a policy reaches  $U$ . We still need to prove that no policy can exceed  $U$ .

## The greedy policy attains the fixed point

Choose one maximizing action at each state:

$$\pi_g(x; U) \in \operatorname{argmax}_a \left[ r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) U(x') \right].$$

Fix these choices and write  $\pi_g = \pi_g(U)$ . Then

$$T^{\pi_g(U)} U = T^* U = U \quad \Longrightarrow \quad \boxed{V^{\pi_g(U)} = U.}$$

The policy is:

- **deterministic**: one chosen action at each state;
- **Markov**: it uses only the current state;
- **stationary**: the rule does not change with time.

## Approximate fixed points and Bellman residuals

Iteration approaches the fixed point, but we stop after finitely many updates. Since  $U$  is unknown, we measure how much one update changes a candidate  $V$ :

$$\text{BR}^*(V) = T^*V - V, \quad \text{BR}^\pi(V) = T^\pi V - V.$$

These are **Bellman residuals**; their norms are **Bellman errors**. They are computable from the candidate and the known model.

$$\|V - U\|_\infty \leq \frac{\|T^*V - V\|_\infty}{1 - \gamma}.$$

A small residual guarantees a small distance to the fixed point.

For example, with  $\gamma = 0.9$ , a residual norm at most 0.02 guarantees value error at most 0.2.

## The residual error bound: proof

Let  $T$  be any of the four operators, and  $Z$  its fixed point. For a candidate  $F$  on the same domain, add and subtract  $TF$ :

$$\begin{aligned}\|F - Z\|_{\infty} &= \|F - TF + TF - TZ\|_{\infty} \\ &\leq \|F - TF\|_{\infty} + \|TF - TZ\|_{\infty} \\ &\leq \|F - TF\|_{\infty} + \gamma\|F - Z\|_{\infty}.\end{aligned}$$

Rearrange and divide by  $1 - \gamma > 0$ :

$$\boxed{\|F - Z\|_{\infty} \leq \frac{\|TF - F\|_{\infty}}{1 - \gamma}.$$

The proof uses the fixed-point identity, the triangle inequality, and contraction.

## Policy error bounds and a stopping rule

The same proof gives, for every stationary Markov policy,

$$\|V - V^\pi\|_\infty \leq \frac{\|T^\pi V - V\|_\infty}{1 - \gamma}, \quad \|Q - Q^\pi\|_\infty \leq \frac{\|T^\pi Q - Q\|_\infty}{1 - \gamma}.$$

For  $T^*$  on action values, the bound likewise controls distance to its unique fixed point.

Update every entry using the same current table:  $V_{k+1} = T^\pi V_k$ . Then  $V_{k+1} - V_k$  is the residual of  $V_k$ , so

$$\|V_{k+1} - V_k\|_\infty \leq (1 - \gamma)\varepsilon \implies \|V_k - V^\pi\|_\infty \leq \varepsilon.$$

This certifies value error. Next, we compare  $U$  with every policy's return.

# Why the Fixed Point Is the Optimal Value



## Comparing with every stationary policy

We know  $V^{\pi_g} = U$ . Could another policy do better? First consider all stationary Markov policies  $\Pi_{\text{stat}}$ , including randomized ones. For any  $\pi \in \Pi_{\text{stat}}$ , an average is at most the maximum:

$$T^\pi U \leq T^* U = U.$$

Apply  $T^\pi$  to both sides. Monotonicity preserves the inequality:

$$(T^\pi)^2 U \leq T^\pi U \leq U.$$

Repeating this argument gives, for every integer  $k \geq 1$ ,

$$(T^\pi)^k U \leq U.$$

The superscript  $k$  means repeated application of the operator, not an ordinary scalar power. Each inequality holds at every state.

## Taking the limit proves stationary optimality

Policy contraction gives

$$(T^\pi)^k U \longrightarrow V^\pi.$$

Since each iterate is at most  $U$ , take the limit at each state:

$$V^\pi \leq U \quad \text{for every } \pi \in \Pi_{\text{stat}}.$$

The greedy policy attains  $U$ , so the two bounds meet:

$$U(x) = V^{\pi_g(U)}(x) = \sup_{\pi \in \Pi_{\text{stat}}} V^\pi(x).$$

**Remaining question:** could using time, past states, or past rewards improve the return?

A history-dependent policy need not have a single stationary operator  $T^\pi$ . We will start again from  $U = T^*U$  and compare returns after each observed history.

## The policies we must compare against

Before the action at time  $t$ , the observed history is

$$H_t = (X_1, A_1, R_1, \dots, X_{t-1}, A_{t-1}, R_{t-1}, X_t), \quad H_1 = X_1.$$

An admissible policy  $\pi = (\pi_t)_{t \geq 1}$  chooses

$$\pi_t(a \mid H_t) \geq 0, \quad \sum_{a \in \mathcal{A}} \pi_t(a \mid H_t) = 1.$$

It may randomize, depend on time, and use the entire observed history. It cannot use future information.

Let  $\Pi_{\text{all}}$  denote this class. Initialize  $X_1 = x$ , then define

$$V^\pi(x) := \mathbb{E}_x^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} R_t \right].$$

$\mathbb{E}_x^\pi$  means expectation under this initialization and policy. For such a policy, the stationary Bellman equation need not apply.

## The one-step bound after a history

Since  $U = T^*U$ , each available action satisfies

$$U(x) \geq r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) U(x').$$

The joint Markov assumption gives

$$\begin{aligned}\mathbb{E}[R_t \mid H_t, A_t] &= r(X_t, A_t), \\ \mathbb{E}[U(X_{t+1}) \mid H_t, A_t] &= \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid X_t, A_t) U(x').\end{aligned}$$

Substitute these into the actionwise bound:

$$\mathbb{E}[R_t + \gamma U(X_{t+1}) \mid H_t, A_t] \leq U(X_t).$$

Linearity of expectation needs no conditional independence between  $R_t$  and  $X_{t+1}$ .

## Averaging over any action choice

Fix any  $\pi \in \Pi_{\text{all}}$ . Given  $H_t$ , average over its action:

$$\begin{aligned} & \mathbb{E}^{\pi}[R_t + \gamma U(X_{t+1}) \mid H_t] \\ &= \sum_{a \in \mathcal{A}} \pi_t(a \mid H_t) \left[ r(X_t, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid X_t, a) U(x') \right] \\ &\leq \sum_{a \in \mathcal{A}} \pi_t(a \mid H_t) U(X_t) = U(X_t). \end{aligned}$$

Every action has the same upper bound; any probability mixture does too.

$$\boxed{\mathbb{E}^{\pi}[R_t + \gamma U(X_{t+1}) \mid H_t] \leq U(X_t).}$$

These conditional statements hold almost surely (with probability one). They do not require a history to have positive point probability.

## Unrolling the first two decisions

Fix  $X_1 = x$  and  $\pi \in \Pi_{\text{all}}$ . All expectations below use  $\mathbb{E}_x^\pi$ .

$$\begin{aligned}U(x) &\geq \mathbb{E}_x^\pi[R_1 + \gamma U(X_2)] \\&\geq \mathbb{E}_x^\pi[R_1 + \gamma \mathbb{E}^\pi[R_2 + \gamma U(X_3) \mid H_2]] \\&= \mathbb{E}_x^\pi[R_1 + \gamma R_2 + \gamma^2 U(X_3)].\end{aligned}$$

- The second line substitutes the bound at  $H_2$ ;  $\gamma \geq 0$ .
- $R_1$  is known at  $H_2 = (X_1, A_1, R_1, X_2)$ .
- Tower property: for any integrable random variable  $Y$ ,  $\mathbb{E}_x^\pi[\mathbb{E}^\pi[Y \mid H_2]] = \mathbb{E}_x^\pi[Y]$ .

No stationary recursion for  $V^\pi$  was used.

## The bound for any finite horizon

For every positive integer  $H$ , the claim is

$$U(x) \geq \mathbb{E}_x^\pi \left[ \sum_{t=1}^H \gamma^{t-1} R_t + \gamma^H U(X_{H+1}) \right].$$

**Induction step.** At the next observed history,

$$U(X_{H+1}) \geq \mathbb{E}^\pi [R_{H+1} + \gamma U(X_{H+2}) \mid H_{H+1}].$$

Multiply by  $\gamma^H$ , substitute, and use the tower property:

$$U(x) \geq \mathbb{E}_x^\pi \left[ \sum_{t=1}^{H+1} \gamma^{t-1} R_t + \gamma^{H+1} U(X_{H+2}) \right].$$

$H$  counts decisions;  $H_t$  denotes a random observed history. The case  $H = 1$  starts the induction.

## Both discounted tails vanish

Recall  $|R_t| \leq R_{\max}$ . The finite real vector  $U$  has  $\|U\|_{\infty} < \infty$ . First, the terminal value disappears:

$$\left| \mathbb{E}_x^{\pi} [\gamma^H U(X_{H+1})] \right| \leq \gamma^H \|U\|_{\infty} \rightarrow 0.$$

Second, the partial expected return approaches the full return:

$$\begin{aligned} \left| V^{\pi}(x) - \mathbb{E}_x^{\pi} \left[ \sum_{t=1}^H \gamma^{t-1} R_t \right] \right| &\leq \mathbb{E}_x^{\pi} \left[ \sum_{t=H+1}^{\infty} \gamma^{t-1} |R_t| \right] \\ &\leq \frac{R_{\max} \gamma^H}{1 - \gamma} \rightarrow 0. \end{aligned}$$

Taking the limit in the finite-horizon bound therefore gives

$$\boxed{U(x) \geq V^{\pi}(x) \quad \text{for every } \pi \in \Pi_{\text{all}}.}$$

## The fixed point is the optimal value

Define the supremum (least upper bound) of achievable values:

$$V^*(x) := \sup_{\pi \in \Pi_{\text{all}}} V^\pi(x).$$

We have proved, for every  $x \in \mathcal{X}$ ,

$$\underbrace{V^{\pi_g}(x) = U(x)}_{\text{attainment}} \quad \text{and} \quad \underbrace{V^\pi(x) \leq U(x) \quad \forall \pi \in \Pi_{\text{all}}}_{\text{domination}}.$$

Since  $\pi_g \in \Pi_{\text{all}}$ , these bounds meet:

$$\boxed{V^{\pi_g} = U = V^*}.$$

One deterministic stationary Markov policy is optimal **simultaneously at every initial state**, even against randomized, nonstationary, history-dependent policies.

## What the policy theorem says

**Sufficiency:** an optimal policy can be chosen deterministic, stationary, and Markov. Other optimal policies may also exist.

**A tie example.** One state  $x$ , two actions  $a, b$ :

$$\mathcal{P}(x \mid x, a) = \mathcal{P}(x \mid x, b) = 1, \quad r(x, a) = r(x, b) = 1, \quad \gamma = \frac{1}{2}.$$

Every policy has value

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots = 2.$$

Randomizing or switching actions over time is also optimal here.

The guarantee uses finite, nonempty state/action sets, bounded rewards,  $0 \leq \gamma < 1$ , and a time-homogeneous joint Markov law for reward and next state. It concerns expected discounted return with the state fully observed.

## Action values with an arbitrary continuation

Force the initial state and action to be  $(X_1, A_1) = (x, a)$ , where  $a \in \mathcal{A}$ . From time 2, follow any admissible policy  $\pi$ . Define

$$Q^\pi(x, a) := \mathbb{E}_{x,a}^\pi \left[ R_1 + \sum_{t=2}^{\infty} \gamma^{t-1} R_t \right], \quad Q^*(x, a) := \sup_{\pi \in \Pi_{\text{all}}} Q^\pi(x, a).$$

Here  $\mathbb{E}_{x,a}^\pi$  denotes this forced-first-action experiment. The continuation may use  $H_2 = (x, a, R_1, X_2)$ , and every later history.

**Target:**

$$Q^*(x, a) = r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) V^*(x').$$

We must prove this for all continuations, including ones that remember  $R_1$ .

## Bounding every continuation after the first action

Apply the almost-sure one-step bound repeatedly from time 2, keeping the full history  $H_2$  in the conditioning:

$$\mathbb{E}_{x,a}^{\pi} \left[ \sum_{t=2}^{H+1} \gamma^{t-2} R_t + \gamma^H V^*(X_{H+2}) \mid H_2 \right] \leq V^*(X_2).$$

The deterministic tail bounds also hold conditionally on  $H_2$ . Let  $H \rightarrow \infty$ :

$$\mathbb{E}_{x,a}^{\pi} \left[ \sum_{t=2}^{\infty} \gamma^{t-2} R_t \mid H_2 \right] \leq V^*(X_2) \quad \text{almost surely.}$$

Use the tower property, then the initial reward/transition law:

$$\begin{aligned} Q^{\pi}(x, a) &\leq \mathbb{E}_{x,a}^{\pi} [R_1 + \gamma V^*(X_2)] \\ &= r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) V^*(x'). \end{aligned}$$

## One policy attains every optimal action value

Let  $\pi^* = \pi_g(U)$ , so  $V^{\pi^*} = V^*$  at every state. After the forced first action, use this stationary Markov policy:

$$Q^{\pi^*}(x, a) = r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) V^*(x').$$

It attains the bound for every  $(x, a)$ . Hence

$$Q^*(x, a) = Q^{\pi^*}(x, a).$$

Maximizing the first action recovers the optimal state value:

$$\max_{a \in \mathcal{A}} Q^*(x, a) = (T^* V^*)(x) = V^*(x).$$

In  $Q^*(x, a)$ , the first action may be poor; optimality describes the continuation after that forced action.

## Identifying the action-value fixed points

Recall the policy equation for a stationary Markov  $\pi$ :

$$Q^\pi(x, a) = r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) \sum_{a' \in \mathcal{A}} \pi(a' \mid x') Q^\pi(x', a').$$

This is exactly  $Q^\pi = T^\pi Q^\pi$ .

For the optimum, substitute  $V^*(x') = \max_{a' \in \mathcal{A}} Q^*(x', a')$ :

$$Q^*(x, a) = r(x, a) + \gamma \sum_{x' \in \mathcal{X}} \mathcal{P}(x' \mid x, a) \max_{a' \in \mathcal{A}} Q^*(x', a').$$

This is exactly  $Q^* = T^* Q^*$ .

The policy equation recalls our earlier result. The optimality equation now identifies its unique fixed point with the return-based  $Q^*$ .

# What we have established

## Question

Why do the Bellman equations have unique solutions?

Why do repeated updates converge?

How can we certify an approximate solution?

Why can a deterministic stationary Markov policy be optimal?

## Reason

Each operator is a contraction on a nonempty complete value-table space.

Banach's theorem gives convergence from every initial candidate and a geometric error bound.

The residual norm bounds the value error after division by  $1 - \gamma$ .

A greedy rule attains  $U$ ; conditional return bounds show every admissible policy has value at most  $U$ .

$$V^\pi = T^\pi V^\pi, \quad Q^\pi = T^\pi Q^\pi, \quad V^* = T^* V^*, \quad Q^* = T^* Q^*.$$

The policy equations here apply to stationary Markov  $\pi$ .

## Planning algorithms supported by today's results

### **Iterative policy evaluation**

$V_{k+1} = T^\pi V_k$ , with a fixed policy

### **Value iteration**

$V_{k+1} = T^* V_k$ , followed by greedy action selection

### **Policy iteration**

Evaluate the current policy, then improve it using its value

Today's results give convergence of the first two value sequences. Policy improvement and policy iteration need their own argument.

Next: algorithm steps, worked comparisons, stopping rules, and computational cost.